Section 1.2 — Inequalities and Intervals

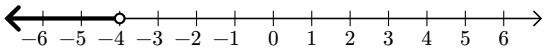
Math 103: College Algebra

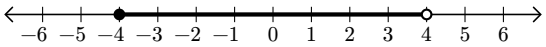
Intervals in the Number Line

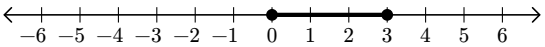
For each interval graphed below:

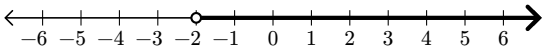
a. Write an inequality that has the solution set shown. Use x as your variable.

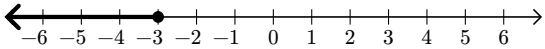
b. Write the set shown in interval notation.

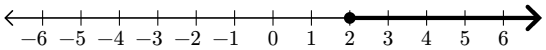
1. 

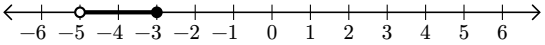
2. 

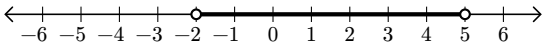
3. 

4. 

5. 

6. 

7. 

8. 

Solving Inequalities

For each inequality below:

a. Solve the inequality.

b. Roughly sketch the solution set on a number line.

c. Write the solution set in interval notation.

Show your work clearly on a separate sheet of paper.

9. $11 - x < 8$
10. $3x + 11 \geq 17$
11. $4x - 7 \leq 11$
12. $3x - 7 > 12$
13. $5x + 2 \leq 3x - 6$
14. $4x - 8 > 5x - 6$
15. $2x - 3 < 11x + 6$
16. $2x + 6 \geq 8x - 7$
17. $-2 < 5x - 7 < 3$
18. $5 > 4x - 7 \geq -7$
19. $3 \leq 6 - x \leq 6$
20. $5 \geq -2x + 5 > -15$

Union and Intersection

Write each union or intersection as a single interval, if possible. Indicate the empty set as $\varnothing$, and the set of all real numbers as $\mathbb{R}$. You may find it helpful to sketch the intervals on a separate sheet of paper.

21. $(-\infty, 14) \cap [-14, \infty)$
22. $(-\infty, 9) \cup [-13, \infty)$
23. $[-6, 0] \cup [-8, \infty)$
24. $(-23, \infty) \cup (-6, 4)$
25. $(-\infty, 8] \cap [-8, \infty)$
26. $[-4, \infty) \cup (-5, 5]$
27. $(-28, -17) \cup (5, 13]$
28. $[-3, 1] \cup (-5, 4)$
29. $[-5, 26] \cup (19, 28]$
30. $(-1, 5) \cap (-5, 1]$
31. $(-1, 16] \cap (1, 10)$
32. $[-5, -4] \cap (-\infty, 9]$
33. $(13, \infty) \cap [-23, \infty)$
34. $[-7, \infty) \cap (-38, 9]$
35. $(-\infty, 45] \cup (-35, \infty)$
36. $[-4, -1) \cap (3, 5)$
37. $(25, \infty) \cup (-68, \infty)$
38. $(-\infty, 8] \cap [-4, 1)$